

Electricity Billing System



CSE347

Section: 04

Summer 2024



Presented By

Mohammed Shawqi Aftab
ID:2022-1-60-013

Jerin Anan Proma
ID: 2022-1-60-132

Fatema Tuz Zannat
ID: 2022-1-60-153

Md. Yousuf Hozaiifa
ID: 2022-1-60-162



Introduction

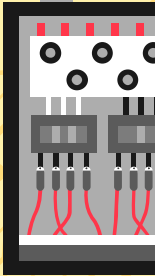
The Electrical Billing System is an advanced software solution designed to automate and modernize electricity billing operations for both individual consumers and commercial entities. Traditionally, billing has been handled manually, leading to inefficiencies, miscalculations and delayed processing. This system eliminates these issues by offering automated billing based on real-time consumption data. It enables users to track their electricity usage, update their billing details and securely pay their bills online. By centralizing billing tasks within a digital framework, the system not only boosts operational accuracy but also enhances user engagement and satisfaction through increased transparency and convenience.

Abstract

This Electrical Billing System is developed to transform the management of electricity consumption and payment processes for households and businesses. The platform offers real-time tracking of energy usage, facilitates the application of dynamic billing rates and ensures that customers receive precise invoices. It allows users to monitor their consumption patterns and manage payments through a user-friendly online interface. The system minimizes human intervention, reducing errors and making the entire billing cycle more streamlined, reliable and accessible to consumers and service providers alike.

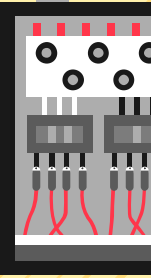
System's Functional Requirements

1. The system shall allow users to login.
2. It shall support two types of users with different functionalities at a same time.
3. The system shall be able to generate bills.
4. It should ask for authentication every time any user try to log in.
5. The system shall verify the payment information before bill transaction.
6. The system shall not support any other language other than English.



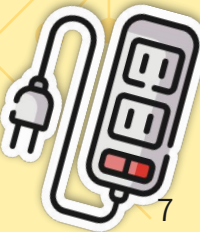
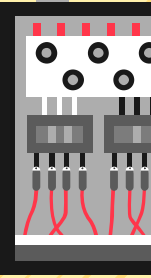
Admin

1. An user logged in as admin shall be able to add new users and modify information of existing users.
2. He/She shall be able to add new meters and edit existing ones.
3. An user logged in as admin shall be able to edit tariff manuals.
4. Admin shall be able to add information of the meter.
5. He/She should be able to generate graphs based on different queries.
6. An user logged in as admin shall be able to search various attributes of the database.



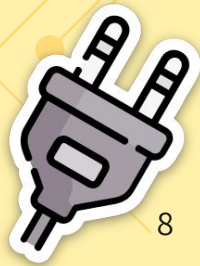
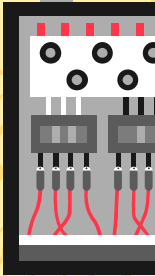
Meter Reader

1. An user logged in as meter reader shall be able to view and search various attributes of the database.
2. He should be able to generate graphs based on different queries.
3. Meter Reader shall be able to update billing information.



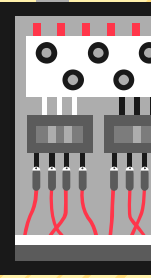
Client

1. An user logged in as client shall be able to see their own bill and other information.
2. They shall be able to download their information to their devices.
3. Clients should be able to see the tariff manual.
4. They shall be able to pay their own bill through the website.
5. An user logged in as client should be able to generate graphs based on different bill related queries.
6. He/She should be able to search bill related information of the database.

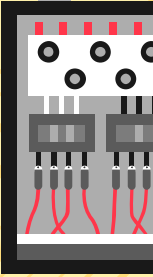


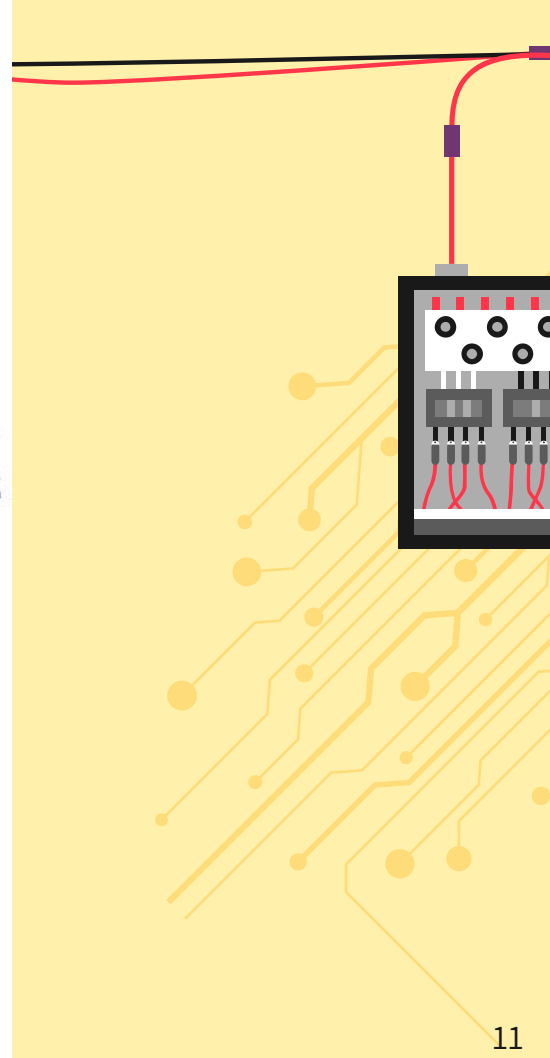
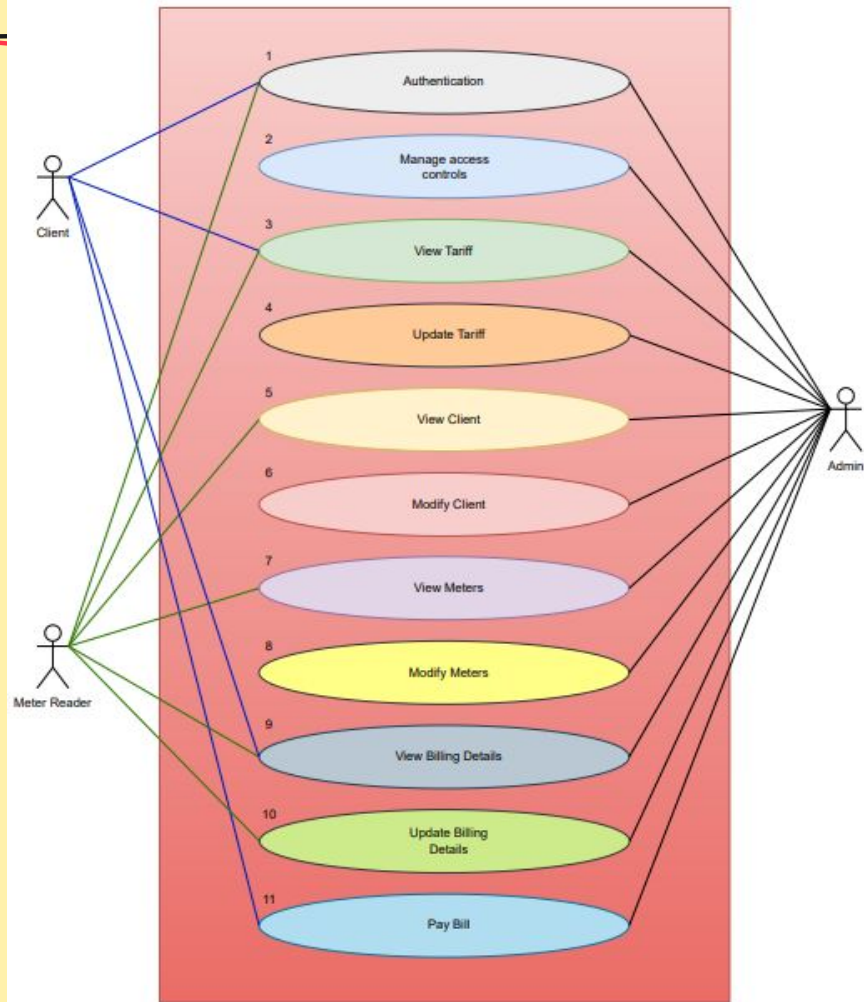
Non Functional Requirements

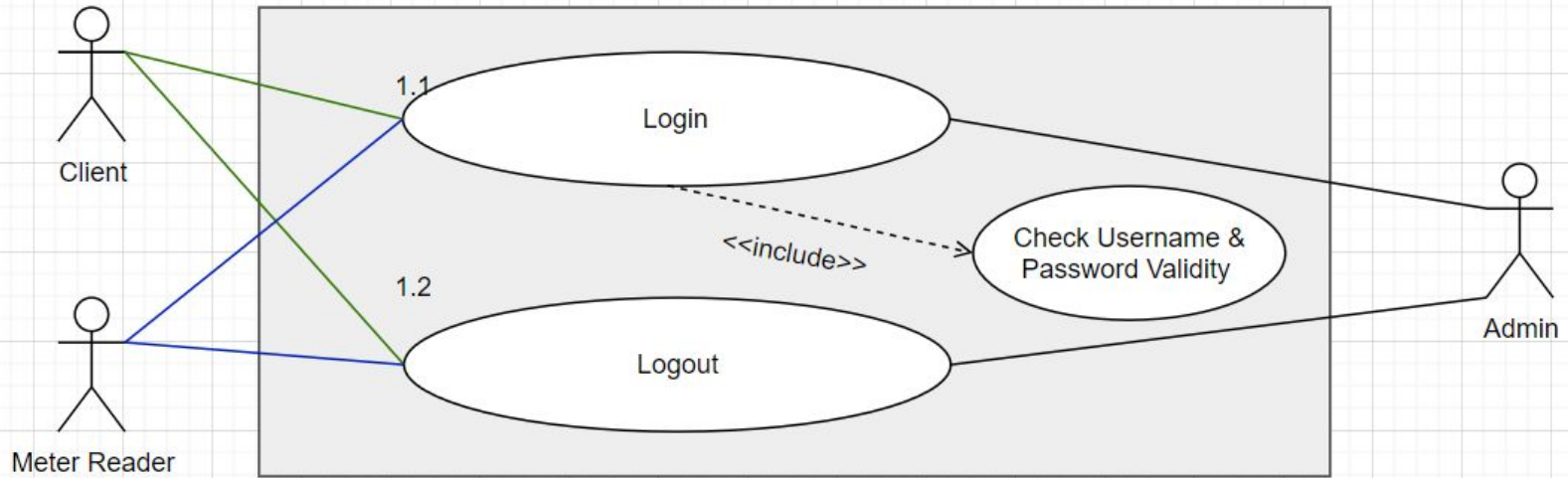
1. The system shall run on web.
2. The passwords shall be encrypted by the system.
3. The system shall be compatible with different devices.
4. It shall work fluently in case of many users and data.
5. The website shall have a user-friendly interface that is easy to navigate.
6. The system shall provide regular backups to prevent data loss.
7. It shall maintain the transactions flawlessly so that no error occurs.
8. The website shall respond to user requests within a specified time to get a smooth experience.

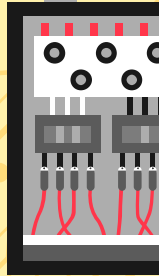
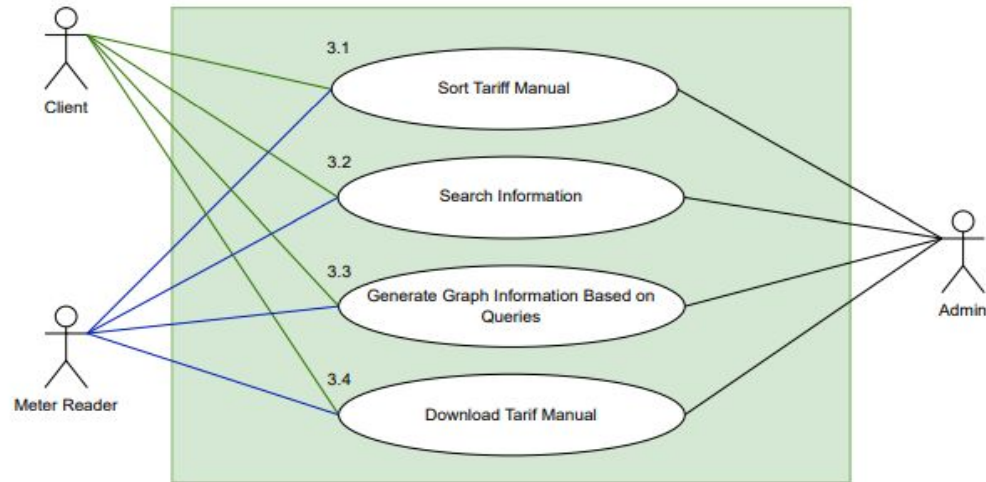
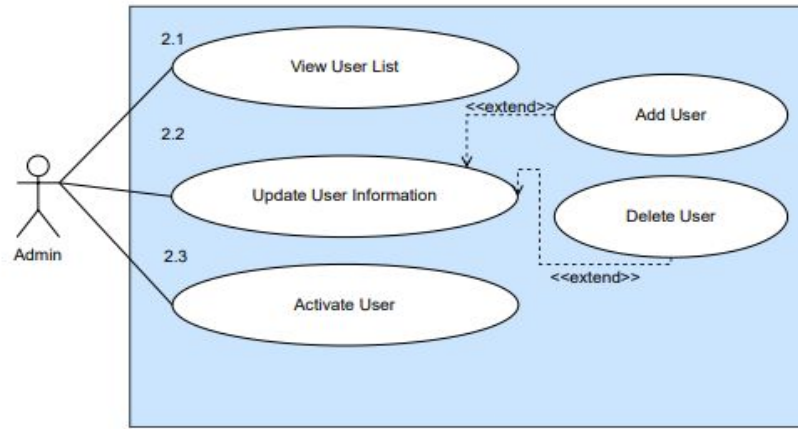


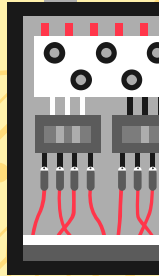
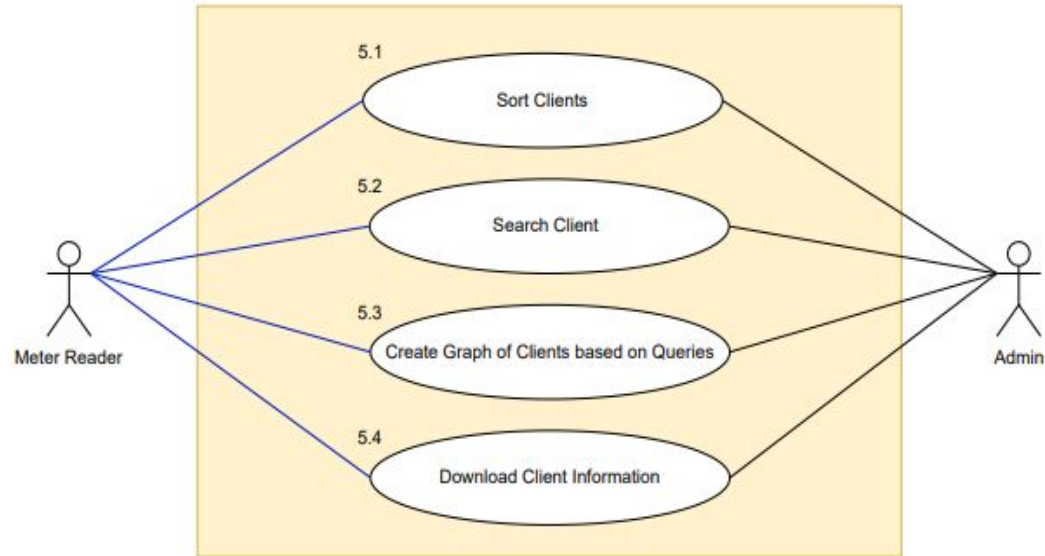
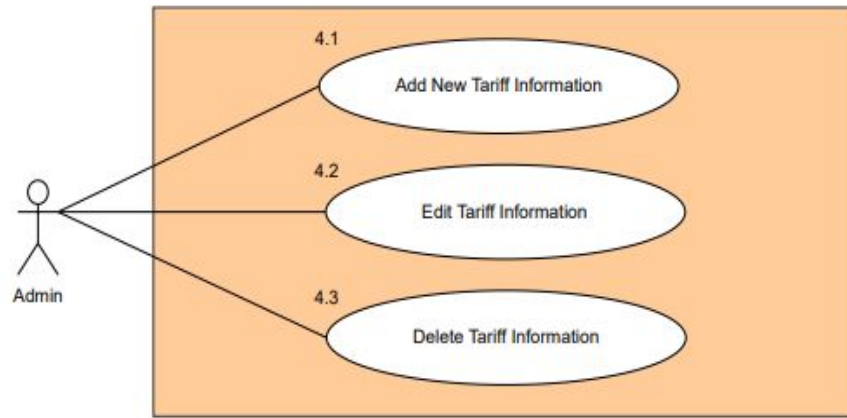
Use Case Diagram

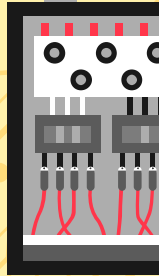
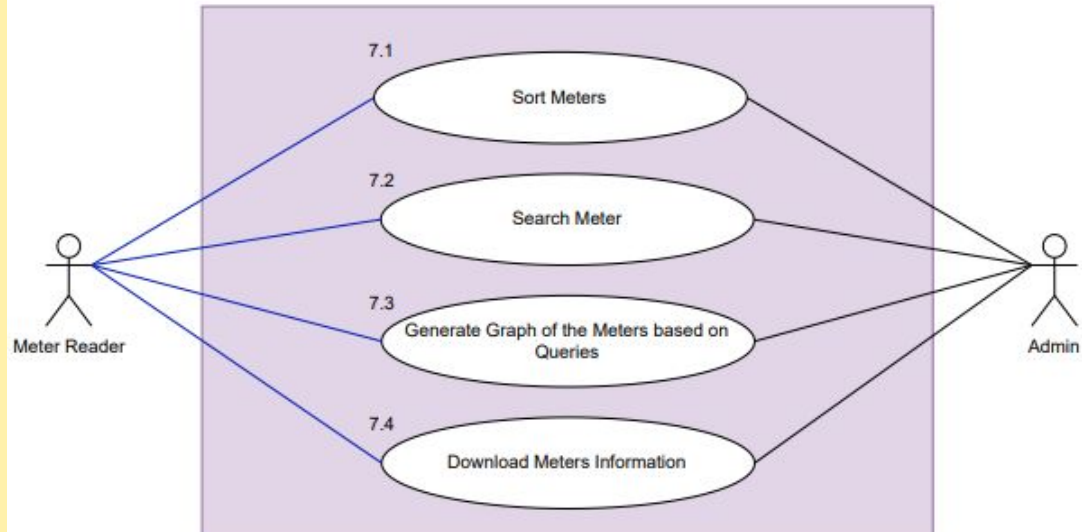
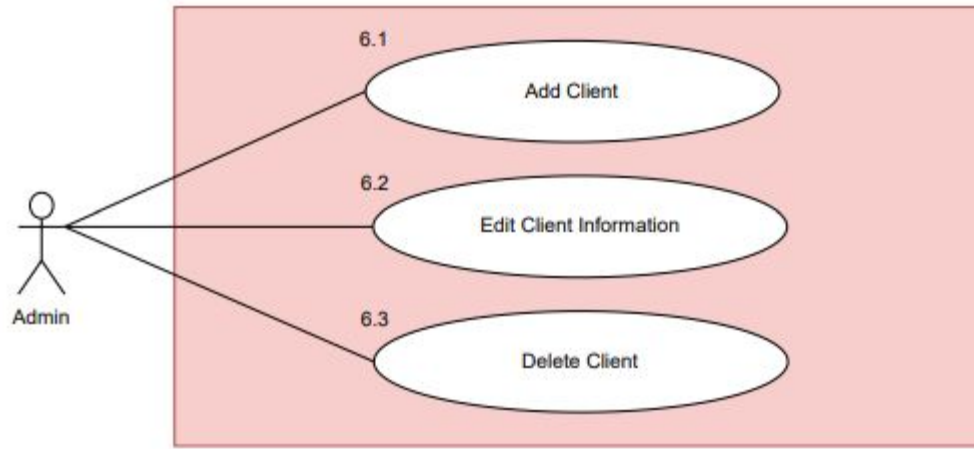


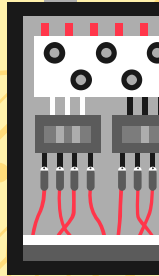
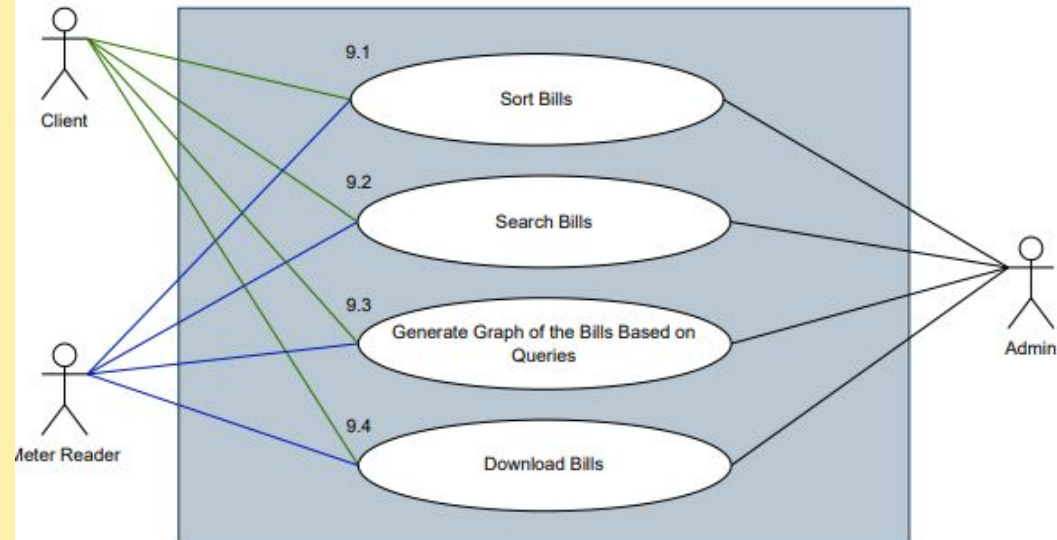
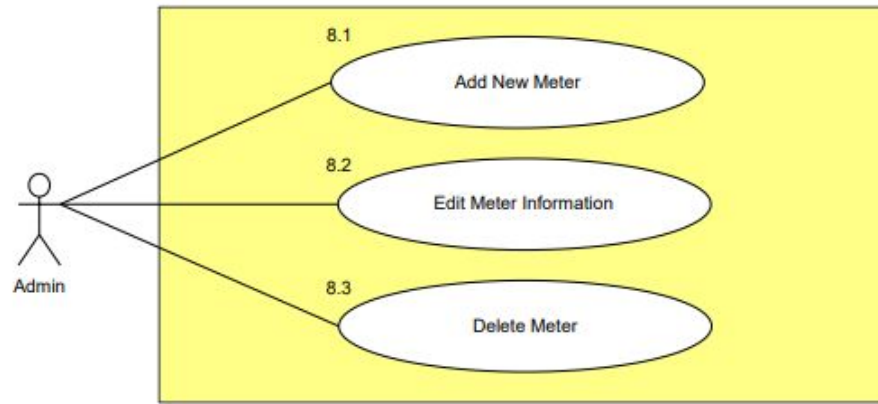


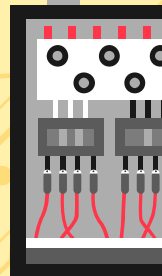
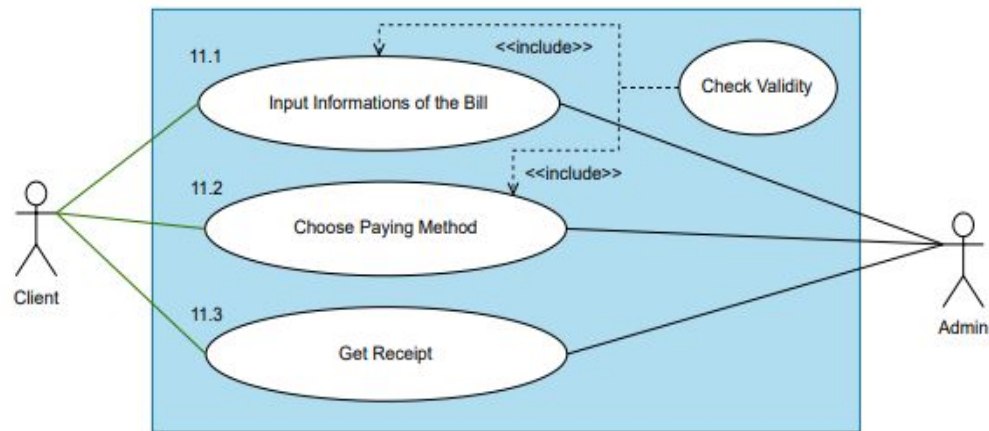
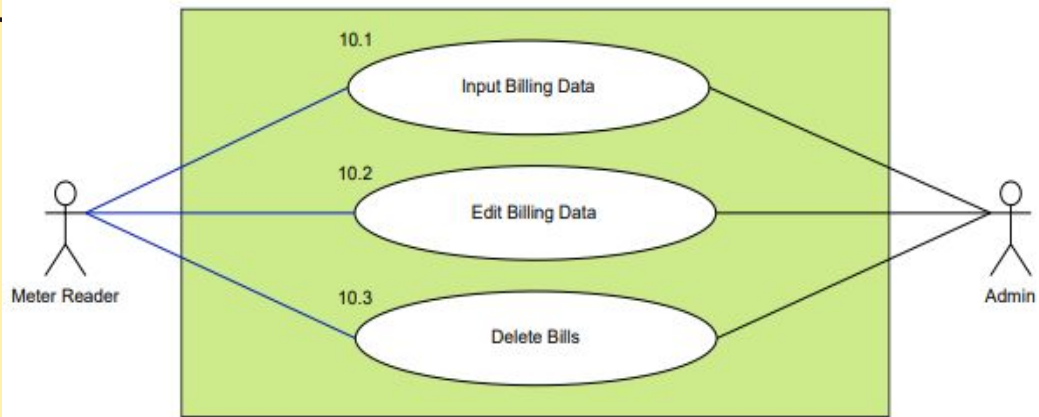


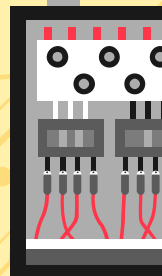
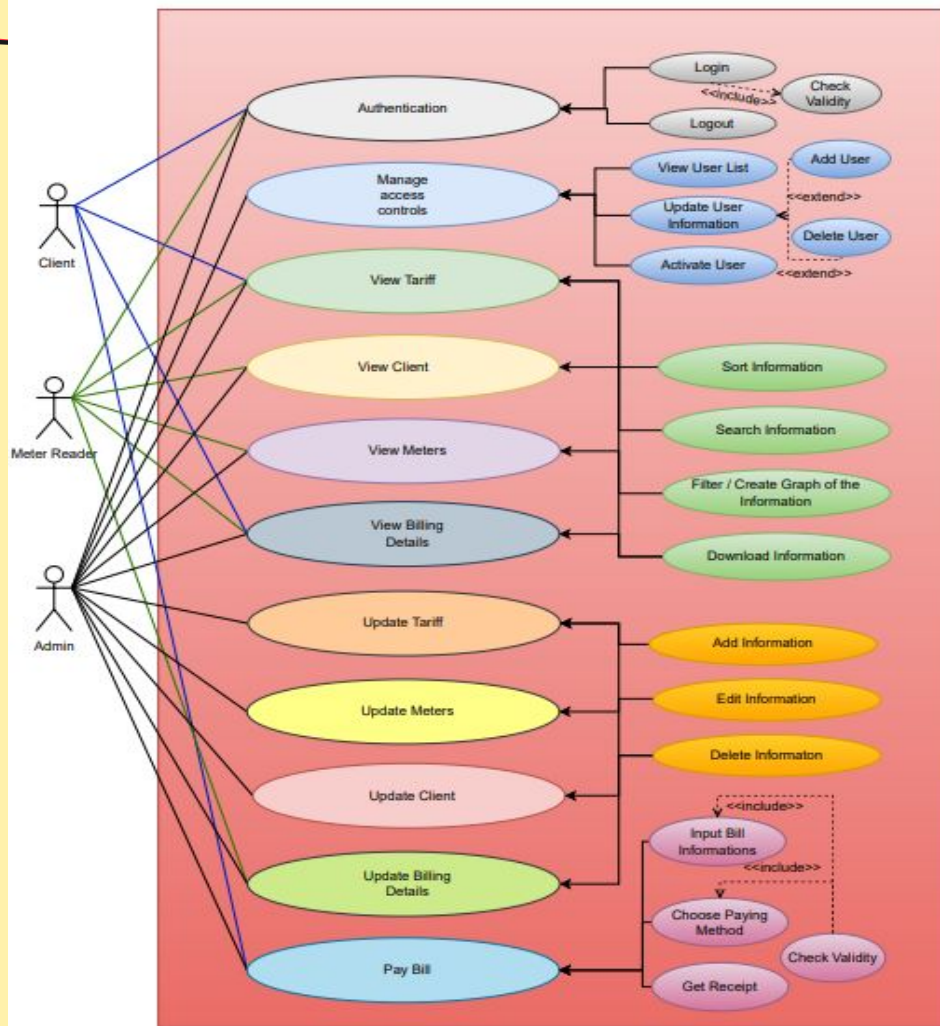




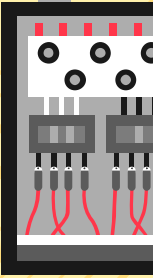




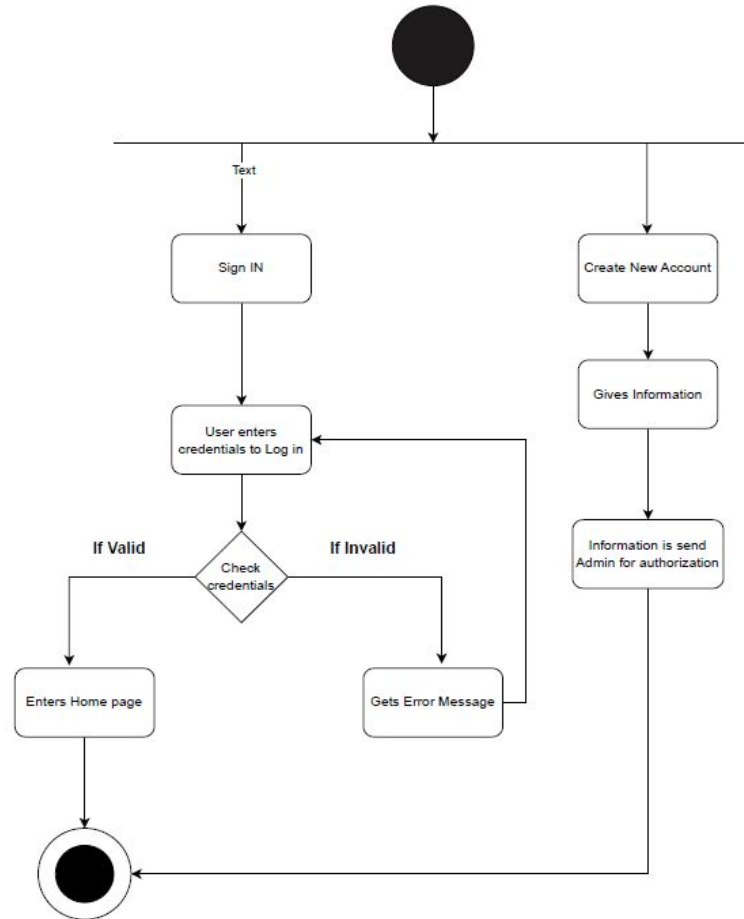




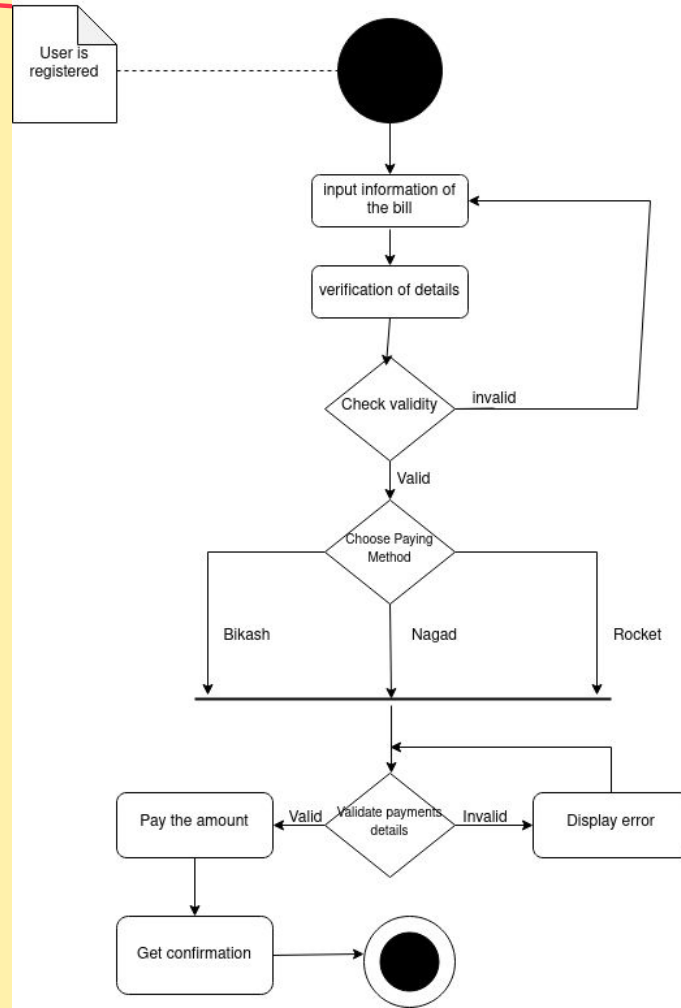
Activity Diagram



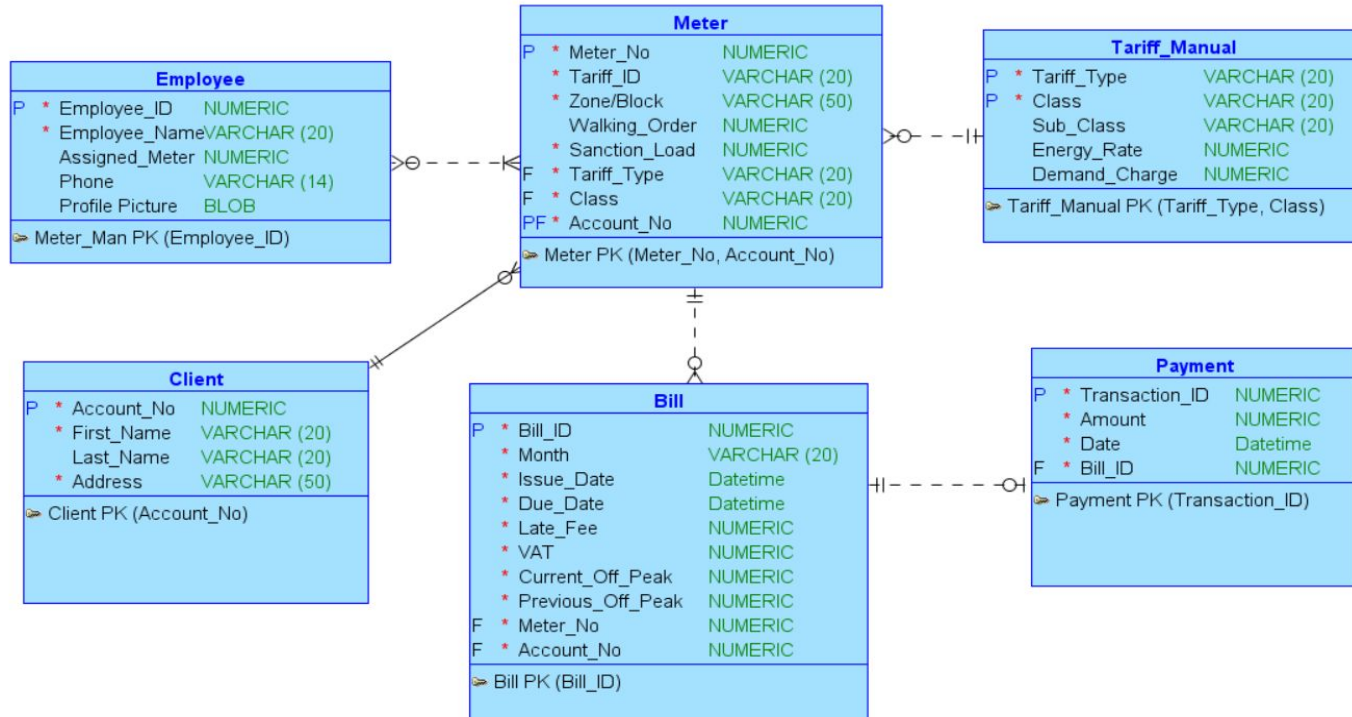
Activity Diagram for Log In



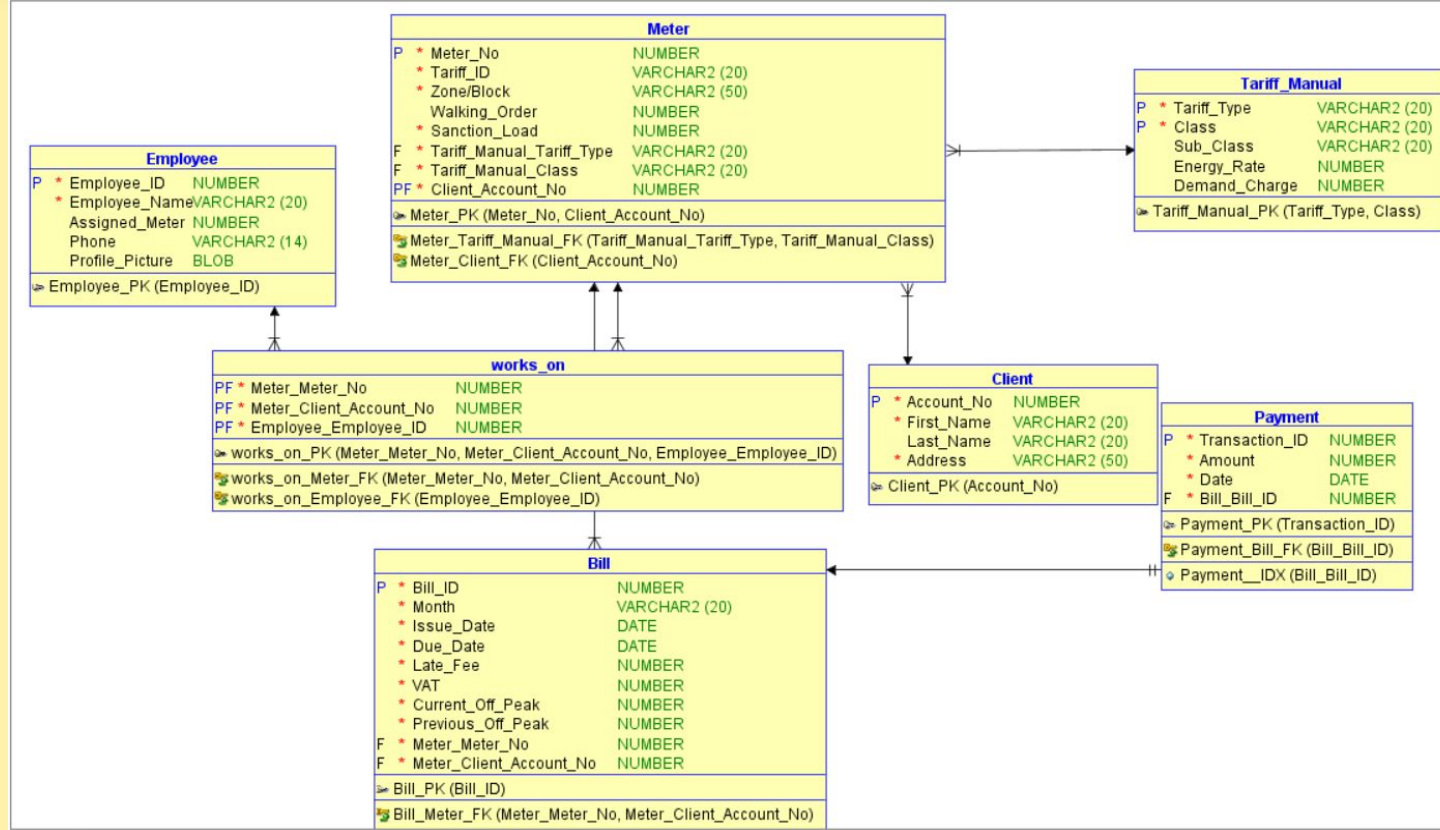
Activity Diagram for Pay Bill



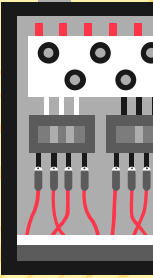
ER Model



Relational Model

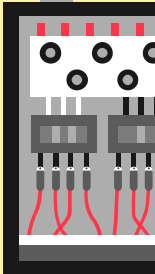


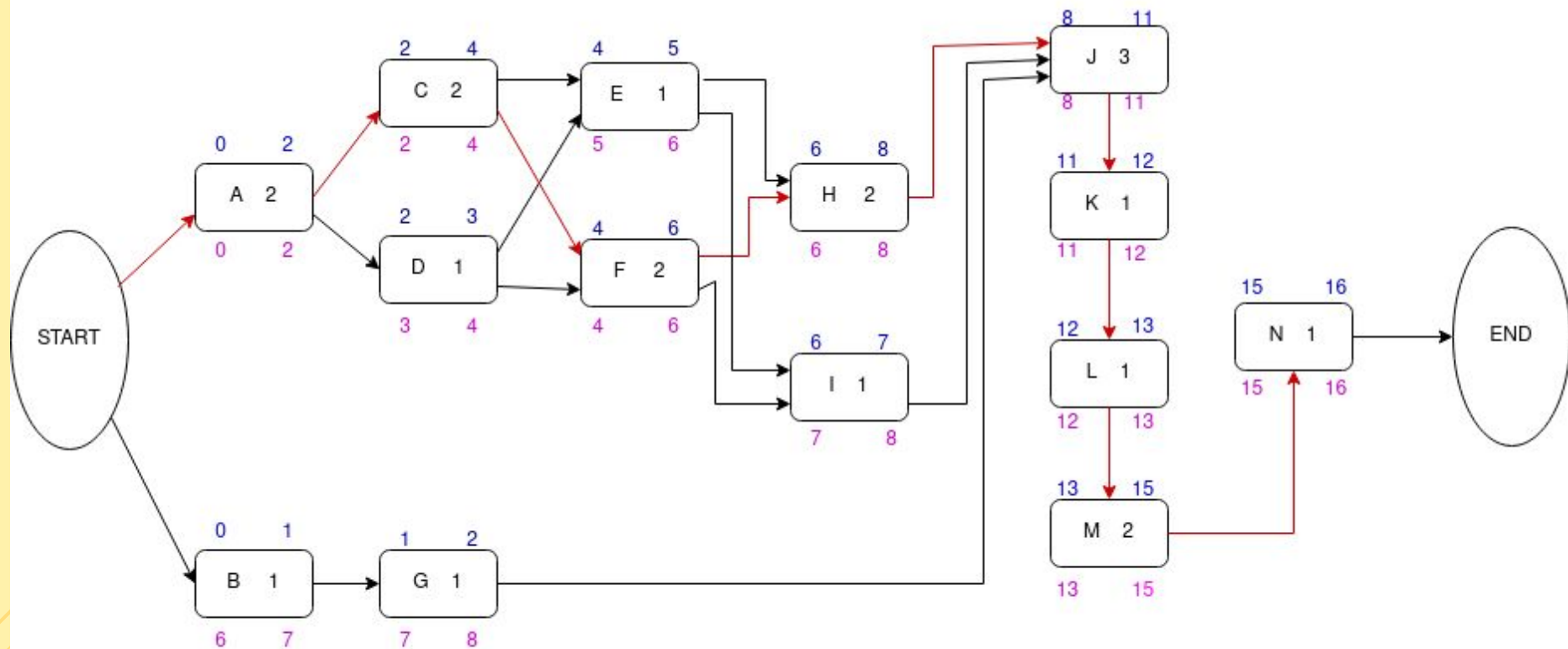
CPM Diagram



Activity Table for Electricity Billing System Project

Activity	Predecessor	Duration (Days)
A - Define System Requirements	None	2
B - Conduct Tariff Manual Analysis	None	1
C - Design Database Schema (UML Diagrams, ERD)	A	2
D - Design User Interface (UI)	A	1
E - Develop Client Module	C, D	1
F - Develop Admin Module	C, D	2
G - Initial Data Collection (Client and Meter Data)	B	1
H - Integrate Meter Module and Payment Processing	E, F	2
I - Develop Bill Generation and Calculation Module	E, F	1
J - Testing and Debugging (Client, Admin, Meter, Payment, Bill Generation, and Tariff Logic)	G, H, I	3
K - Final Review and Approval	J	1
L - User Training and Documentation	K	1
M - System Deployment	L	2
N - Post-deployment Support	M	1





Critical Path: A → C → F → H → J → K → L → M → N

Calculate Floats to Determine Critical Path

Calculating total float of each node:

We know, **Total float of a node** = **LS - ES**

Now,

$$\text{Total float of node A} = 0 - 0 = 0$$

$$\text{Total float of node B} = 6 - 0 = 6$$

$$\text{Total float of node C} = 2 - 2 = 0$$

$$\text{Total float of node D} = 3 - 2 = 1$$

$$\text{Total float of node E} = 5 - 4 = 1$$

$$\text{Total float of node F} = 4 - 4 = 0$$

$$\text{Total float of node G} = 7 - 1 = 6$$

$$\text{Total float of node H} = 6 - 6 = 0$$

$$\text{Total float of node I} = 7 - 6 = 1$$

$$\text{Total float of node J} = 8 - 8 = 0$$

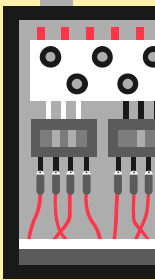
$$\text{Total float of node K} = 11 - 11 = 0$$

$$\text{Total float of node L} = 12 - 12 = 0$$

$$\text{Total float of node M} = 13 - 13 = 0$$

$$\text{Total float of node N} = 15 - 15 = 0$$

A **critical path** in project management is the *longest sequence of activities* that must be finished on time to complete the entire project.



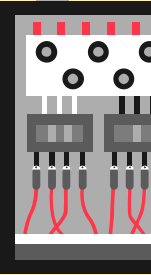
Calculate Floats to Determine Critical Path

From Free Floats-

Critical Path: $A \rightarrow C \rightarrow F \rightarrow H \rightarrow J \rightarrow K \rightarrow L \rightarrow M \rightarrow N$

- **Total Duration= 16 days**

This means that any delay in activities along this path will delay the entire project.



Conclusion

In conclusion, the Electrical Billing System represents a forward-thinking approach to handling electricity billing processes. By incorporating automation and online functionality, the system reduces the reliance on manual operations and minimizes errors associated with traditional billing methods. The ability to access and manage bills in real-time improves customer interaction, giving users more control over their energy consumption and finances. For businesses, this software provides a more efficient and scalable billing solution, contributing to better resource management and customer relations. The system's flexibility guarantees it will continue to be an essential tool for managing energy in the future.

Figma Mockup Link

<https://www.figma.com/proto/CllZCilOqc5qZKm3zrE371/cse347?node-id=6-15&starting-point-node-id=6%3A15&scaling=contain&content-scaling=fixed&show-proto-sidebar=1>



Website Link

https://apex.oracle.com/pls/apex/r/347_project/electric-billing-system/login?session=12281900558214

User Name: **ADMIN**
User Name: **READER**
User Name: **METER**

Password: **ADMIN**
Password: **READER**
Password: **METER**

